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# WaveConvX: Multi-Level Wavelet Enhancement for Histopathology Image Classification

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***Abstract*— Artificial histopathological image classification is emerging as a promising approach for assisting early cancer diagnosis and informing treatment planning. Although convolutional neural network (CNN) based methods have achieved significant success, they still struggle to capture the rich multi-scale and high-frequency details inherent in pathological tissues. To address the above issue, we propose WaveConvX, a novel histopathological image classification model that integrates multi-level wavelet decomposition with the ConvNeXt backbone. Our approach applies a two-stage discrete wavelet transform (DWT) to intermediate feature maps, enabling the decomposition of features into multiple frequency sub-bands. High-frequency components are enhanced using Adaptive Power Gabor Convolution (APGConv), while mid-frequency ranges are refined through tailored attention mechanisms. These processed sub-bands are then fused via inverse wavelet transforms, producing feature maps enriched with both global context and local morphological cues essential for accurate diagnosis. Comprehensive experiments are conducted on three public datasets (BreakHis for breast cancer, KBSMC for gastric cancer, and LC25000 for lung and colon cancer). The results demonstrate that WaveConvX consistently outperforms ten state-of-the-art benchmarks, achieving superior accuracy, F1 scores, and robustness across multiple types and magnifications of cancer. Our work demonstrates the significant potential of wavelet-enhanced CNNs in histopathological image analysis.**

## I. Introduction

Cancer remains one of the leading causes of mortality worldwide, with histopathological examination of tissue slides serving as the gold standard for diagnosis [1], [2]. While this traditional approach provides critical insights for treatment planning, manual analysis by pathologists faces significant limitations including subjective interpretation, inter-observer variability, and time constraints given the large slide sizes and subtle tissue features. Deep learning-based methods have demonstrated promising advances for automating histopathological image analysis [3], [4], yet standard convolutional neural networks (CNNs) struggle to fully capture the inherent multi-scale nature and frequency-rich attributes of pathological tissues [5]–[7].

Conventional CNN architectures primarily rely on hierarchical feature extraction through fixed-size convolutional filters, which often fail to simultaneously integrate both global tissue organization and local cellular morphological cues crucial for accurate diagnosis [8]. Frequency domain representations offer a principled approach to decompose histopathological images into multiple sub-bands, revealing tissue structures at various scales [9]–[11]. Previous attempts to incorporate wavelet transformations with CNN pipelines have shown potential for capturing textural details essential for malignancy detection [12]–[14].

We propose **WaveConvX**, a novel model that integrates *multi-level* wavelet decomposition, trainable frequency-domain operations, and a powerful ConvNeXt backbone to address the multi-scale classification challenges in histopathology. Our *Multi-Level Wavelet Enhancement Module* (1) performs two-stage wavelet transforms on intermediate feature maps, (2) processes each frequency band with specialized blocks tailored to specific histopathological patterns, and (3) reconstructs the enhanced features via inverse wavelet transforms. The refined feature representations capture both fine-grained cellular details and global tissue architecture simultaneously.

We evaluate our approach on three diverse histopathological cancer datasets, demonstrating that WaveConvX consistently outperforms plain ConvNeXt and ten other state-of-the-art baselines, including classical CNNs, transformer-based models, and wavelet-token mixing methods. Our comprehensive ablation studies confirm that targeted processing of specific frequency ranges offers consistent performance gains.

**Key contributions** of this work include:

- We propose a novel multi-level wavelet enhancement module that inserts two discrete wavelet transforms at an intermediate stage in ConvNeXt, capturing distinct frequencies in pathology features.
- We introduce a set of tailored sub-band processing layers: Adaptive Power Gabor Convolution (APGConv) for high-frequency signals (e.g., nuclear boundaries), self-attention for mid-high frequency channels, and channel attention for mid-low frequency components.
- We conducted comprehensive experiments on three publicly available datasets and compared ten contemporary baselines, including wavelet-based and transformer-based models, and experimentally demonstrated that our model performed better.
- We perform thorough ablation studies demonstrating how wavelet families, wavelet placement, sub-band processing methods, and data augmentation interact to produce robust classification results.

## II. Related Work

### A. Wavelet Transform in Medical Image Analysis

Wavelet transforms have long served as a robust approach for multi-resolution analysis, decomposing signals

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or images into localized frequency sub-bands [9]–[11]. In breast cancer histopathology, wavelet techniques have been used for texture feature extraction [15], denoising [16], and region segmentation. Early methods focused on hand-crafted wavelet feature descriptors [12], [17], but recent works have combined wavelets with deep neural networks to improve representation learning [12]–[14]. Many such efforts, however, apply wavelet transforms only once or treat wavelet decomposition as a mere input preprocessing step. In contrast, our approach embeds two levels of wavelet decomposition *within* the CNN feature hierarchy and applies frequency-specific enhancements before inverse reconstruction.

### *B. CNNs for Histopathological Images*

Deep CNNs have become mainstays in medical image classification. Transfer learning from ImageNet-pretrained models (VGG, ResNet, DenseNet, MobileNet, EfficientNet, etc.) has proven especially valuable for tasks such as breast cancer histopathology [18], [19]. Nonetheless, pathological images contain subtle morphological cues spanning multiple scales [8], [20], and naive CNN-based pipelines may lose some high-frequency detail during aggressive downsampling. Our WaveConvX leverages wavelet sub-bands to preserve these detailed features while still benefiting from a modern CNN backbone.

### *C. Modern Architectures and WaveMix*

Recent research has explored hybrid or purely transformer-based designs, including the Vision Transformer (ViT) [21] and Swin Transformer [22], for medical imaging. However, the self-attention mechanism alone does not inherently address multi-frequency representation. ConvNeXt [23] adopts design insights from Transformers while retaining convolutional inductive biases. Another noteworthy approach is WaveMix [24], which uses wavelet-based token mixing to achieve competitive image classification performance. Here, we systematically compare against these modern architectures, showing that our multi-stage wavelet enhancement can yield higher accuracy for histopathological tasks.

## III. METHODOLOGY

In this section, we present our **WaveConvX** architecture for histopathological image classification. As illustrated in Fig. 1, our approach combines ConvNeXt [23] backbone with a novel frequency-domain enhancement mechanism. We first outline the complete pipeline (§III-A), then describe our multi-level wavelet decomposition strategy (§III-B), and finally detail the specialized frequency-specific processing techniques (§III-C), with particular focus on our innovative Adaptive Power Gabor Convolution for high-frequency feature enhancement.

### *A. Overview of WaveConvX*

Our end-to-end pipeline consists of four interconnected modules:

1) **Augmentation**: Input histopathology images undergo rotation, random noise addition, and other transformations to enhance model robustness and generalization.
2) **ConvNeXt Feature Extraction**: The augmented images are processed through the initial stages (1-3) of a ConvNeXt-Base [23] backbone, generating intermediate feature maps $f_{\text{inter}} \in \mathbb{R}^{C\times H\times W}$ that capture hierarchical visual patterns.
3) **MultiWavelet Enhance**: This novel module—our primary contribution—transforms $f_{\text{inter}}$ into the wavelet domain, applying specialized processing to different frequency components before reconstruction. The module outputs enhanced features $f_{\text{enh}}$ that better represent histopathological structures at multiple scales.
4) **ConvNeXt Classification**: The enhanced features flow through the remaining ConvNeXt stages (4-6), followed by global average pooling and softmax classification.

The critical innovation in our approach is the **MultiWavelet Enhance** module (depicted in detail at the bottom of Fig. 1), which decomposes features into frequency sub-bands that align with different histopathological structures—cell boundaries and nuclear details in high frequencies, tissue organization in low frequencies, and transitional structures in mid-frequencies.

### *B. Multi-Level Wavelet Decomposition and Reconstruction*

Let $X \in \mathbb{R}^{C\times H\times W}$ represent the intermediate feature maps entering our wavelet module. As shown in the left portion of the MultiWavelet Enhance module in Fig. 1, we apply a two-level Haar-based Discrete Wavelet Transform (DWT):

$$\text{DWT}(X) = \left[LL_1,\, LH_1,\, HL_1,\, HH_1\right] \tag{1}$$

where $LL_1$ contains low-frequency information (approximation), while $LH_1$, $HL_1$, and $HH_1$ represent horizontal, vertical, and diagonal high-frequency details, respectively. We further decompose the $LL_1$ component:

$$\text{DWT}(LL_1) = \left[LL_2,\, LH_2,\, HL_2,\, HH_2\right] \tag{2}$$

After applying frequency-specific processing (described in §III-C), we perform reconstruction via inverse DWT (IDWT), as shown in the right portion of the module:

$$LL_1' = \text{IDWT}\left[\, LL_2',\ LH_2',\ HL_2',\ HH_2'\right] \tag{3}$$

$$X' = \text{IDWT}\left[\, LL_1',\ LH_1',\ HL_1',\ HH_1'\right] \tag{4}$$

Finally, we add the original input as a residual connection to preserve feature integrity:

$$\tilde{X} = X + X' \tag{5}$$

This residual addition ensures that both original and enhanced features contribute to the final representation $\tilde{X}$, which is then fed back into the main ConvNeXt pipeline.

We implement DWT operations efficiently using $2 \times 2$ stride-2 convolutions with fixed Haar wavelet kernels, and IDWT using $2 \times 2$ stride-2 transposed convolutions. Each operation is grouped by channel count to ensure independent transformation per channel with minimal computational overhead.



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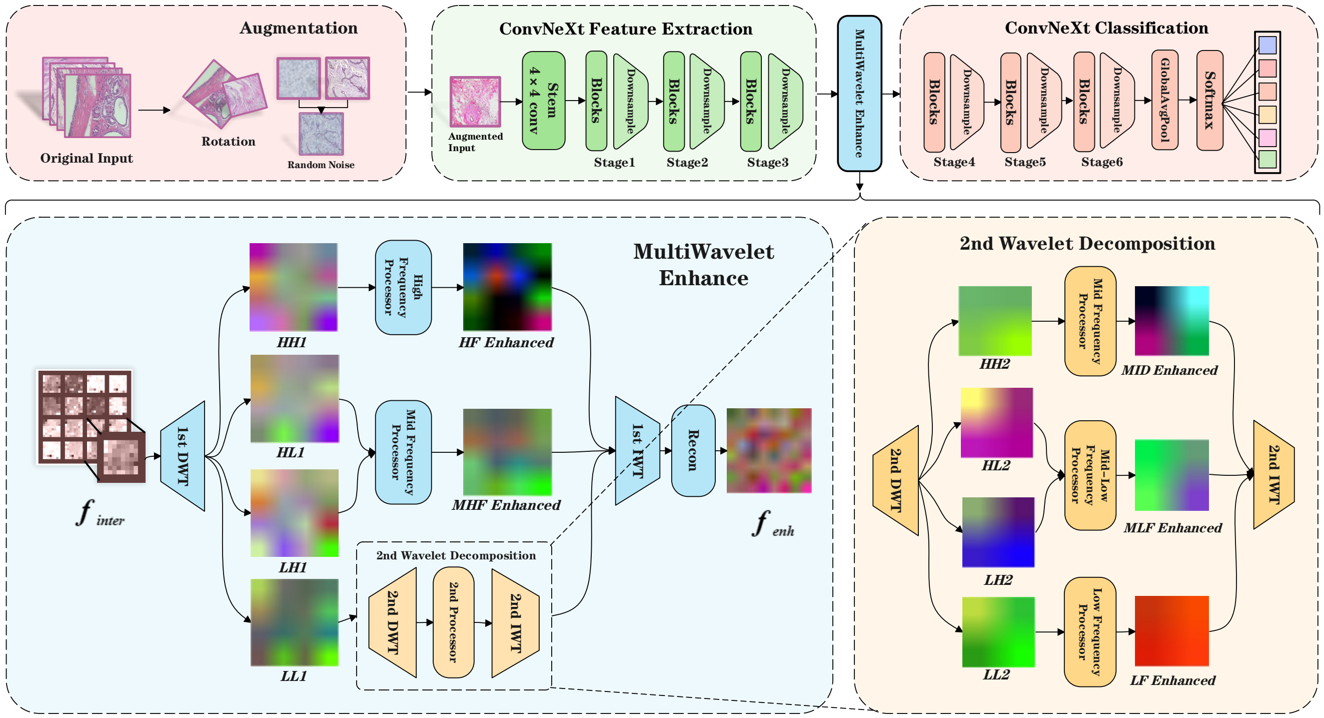


Fig. 1: The WaveConvX framework includes four modules: (1) Input augmentation, (2) ConvNeXt feature extraction, (3) MultiWavelet Enhance (Our primary contribution), and (4) ConvNeXt Classification for final prediction. The MultiWavelet Enhance module uses two-level wavelet decomposition to split features into frequency bands, applies specialized processing to each band, and reconstructs enhanced features via inverse wavelet transforms.

### C. *Frequency-Specific Processing*

As illustrated in Fig. 2, we apply specialized processing to different frequency sub-bands to target specific histopathological patterns:

*1) High-Frequency (HF) Sub-band: Adaptive Power Gabor Convolution:* The $HH_1$ sub-band captures fine details critical for cancer cell detection, including cell boundaries and nuclear texture. To enhance these features, we developed the **Adaptive Power Gabor Convolution** (APGConv), our key innovation illustrated in Fig. 3.

APGConv enhances traditional Gabor filters with a learnable fractional exponent, allowing adaptive feature extraction. For each output channel $k$, we define a Gabor function on a $5 \times 5$ grid:

$$g_k(x, y) = \exp\Big(-\frac{x_\theta^2 + \gamma_k^2\, y_\theta^2}{2\,\sigma_k^2}\Big)\ \cos\Big(2\,\pi\,\frac{x_\theta}{\lambda_k}\Big) \tag{6}$$

where the rotated coordinates are:

$$\begin{aligned} x_\theta &= x\cos(\theta_k) + y\sin(\theta_k) \\ y_\theta &= -x\sin(\theta_k) + y\cos(\theta_k) \end{aligned} \tag{7}$$

The parameters $\theta_k$ (orientation), $\lambda_k$ (wavelength), $\sigma_k$ (scale), and $\gamma_k$ (aspect ratio) are all learnable, enabling adaptive adjustment to different tissue patterns.

Our key innovation is the introduction of a trainable exponent $\alpha_k$ that modulates filter response:

$$g'_k(x, y) \;=\; \operatorname{sign}\big(g_k(x, y)\big)\,\Big|g_k(x, y)\Big|^{\alpha_k} \tag{8}$$

When $\alpha_k < 1$, this compresses large values while expanding small ones, allowing the filter to adapt to subtle texture variations. When $\alpha_k > 1$, stronger signals are emphasized, providing flexibility in feature detection.

The final convolution combines the fractional Gabor kernel with learnable weights:

$$\widetilde{W}_k \;=\; W_k \,\odot\, g'_k \tag{9}$$

followed by application to the high-frequency sub-band:

$$HH'_1 = \text{Conv2D}\big(HH_1; \widetilde{W}, \text{padding} = 2\big) \tag{10}$$

This is followed by GELU activation and batch normalization, effectively enhancing oriented high-frequency structures crucial for cancer cell detection.

*2) Processing of Other Frequency Bands:* The remaining frequency bands are processed as follows:

- **Mid-High Frequency (MHF):** The $LH_1$ and $HL_1$ sub-bands contain directional edges and mid-level tissue structures. We apply self-attention to capture spatial relationships:

$$\begin{aligned} MHF_1 &= \text{SelfAttn}(LH_1) \\ MHF_2 &= \text{SelfAttn}(HL_1) \end{aligned} \tag{11}$$

  The outputs are averaged to produce $MHF'$, balancing horizontal and vertical detail information.
- **Mid-Frequency (MID):** For the $HH_2$ band, we use a depthwise $3\times3$ convolution followed by a pointwise $1\times$



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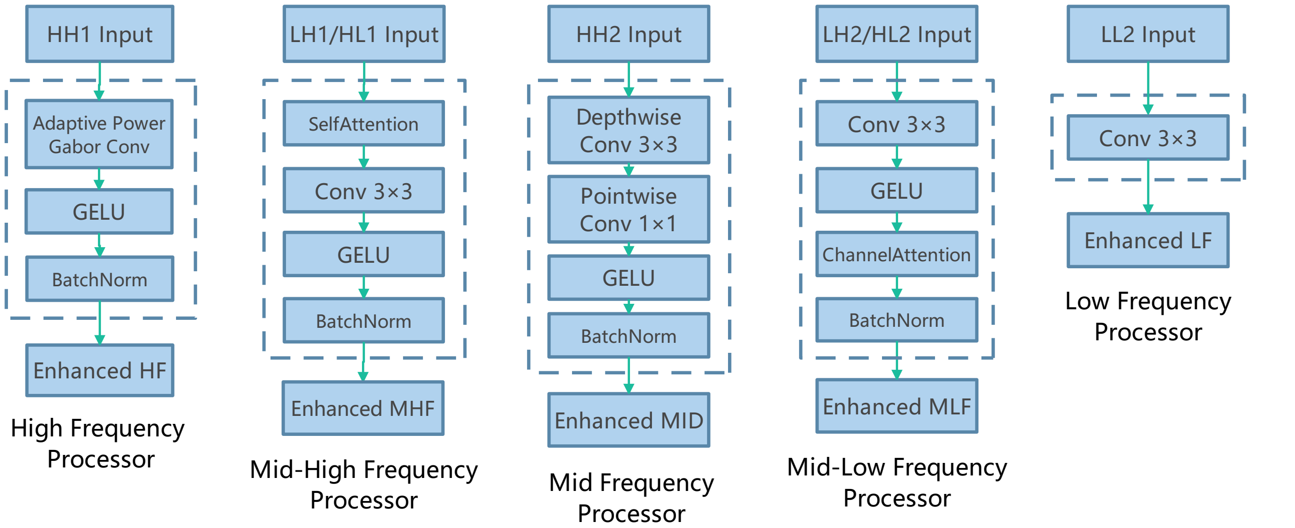


Fig. 2: Frequency-specific processing: HF (HH1) uses APGConv for texture enhancement; MHF (LH1/HL1) applies self-attention for spatial relationships; MID (HH2) uses depthwise+pointwise convolutions; MLF (LH2/HL2) uses channel attention; LF (LL2) uses standard convolutions.

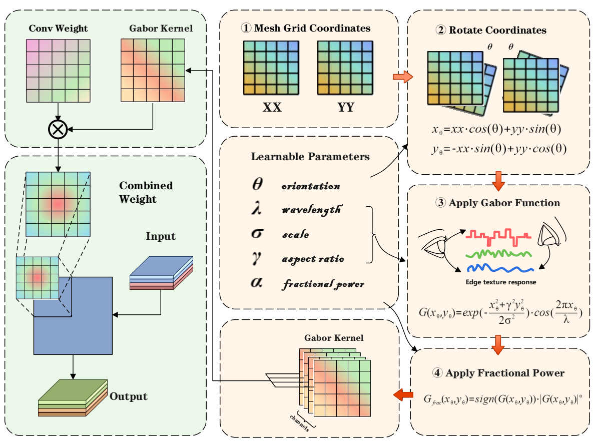


Fig. 3: Illustration of our **Adaptive Power Gabor Convolution (APGConv)** module. The process involves: (1) Creating mesh grid coordinates, (2) Rotating coordinates according to orientation parameter $\theta$, (3) Applying the Gabor function with learnable parameters for wavelength $\lambda$, scale $\sigma$, and aspect ratio $\gamma$, and (4) Applying a fractional power $\alpha$ to adaptively enhance texture details.

1 convolution with GELU activation, capturing diagonal patterns efficiently.

- **Mid-Low Frequency (MLF):** The $LH_2$ and $HL_2$ bands are processed with channel attention that adaptively weights feature channels, followed by convolution, GELU, and batch normalization.
- **Low-Frequency (LF):** We apply a standard $3 \times 3$ convolution to $LL_2$ to refine coarse structural features representing overall tissue organization.

After processing each frequency band, we reconstruct the signal via inverse wavelet transforms following Equations 3-5. This reconstruction, with the residual connection, ensures that original feature information is preserved while being enhanced by our frequency-specific processing, creating a feature representation that captures multi-scale tissue characteristics essential for accurate histopathological classification.

## IV. Experiments and Results

### A. Experimental Setup

We evaluate WaveConvX on three diverse histopathological datasets:

- **BreakHis** [24]: 7,909 breast cancer images at multiple magnifications (40×, 100×, 200×, 400×) with binary (benign/malignant) and 8-class subtypes.
- **KBSMC** [25]: 5,000 gastric cancer images (512×512 pixels) with 5 cancer progression categories.
- **LC25000** [26]: 25,000 lung and colon cancer images (256×256 pixels) across 5 classes.

We use standard train-validation-test splits (70%-10%-20%) across all datasets. For implementation, we build on ConvNeXt-Base [23] pretrained on ImageNet, with our multi-level wavelet enhancement module inserted after Stage 3 of the backbone. Models are trained using AdamW optimizer (lr=$10^{-4}$) for 50-70 epochs with appropriate loss functions (BCEWithLogitsLoss for binary, cross-entropy for multi-class) and robust data augmentation.

### B. Comparative Analysis

We benchmark WaveConvX against ten state-of-the-art models spanning traditional CNNs (VGG16, ResNet-50, DenseNet-121, MobileNetV3, EfficientNet-B4, RegNetY-800mf), transformers (ViT-B, Swin-T), hybrid architectures (ConvNeXt-Base), and wavelet-based methods (WaveMix), using accuracy and macro averaged F1-scores.

Table I shows that for binary classification on BreakHis, WaveConvX outperforms all benchmarks across all magnifications, achieving 99.6% F1-score at 40× and 99.0% at 400×,



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TABLE I: Performance comparison on BreakHis dataset for both binary classification (benign vs. malignant) and multi-class (8-class) classification across different magnifications.

| Model | Binary Classification | | | | | | | | 8-Class Classification | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 40× | | 100× | | 200× | | 400× | | 40× | | 100× | | 200× | | 400× | |
| | Acc | F1 | Acc | F1 | Acc | F1 | Acc | F1 | Acc | F1 | Acc | F1 | Acc | F1 | Acc | F1 |
| VGG16 | 96.2 | 95.3 | 97.7 | 97.6 | 96.5 | 95.4 | 90.9 | 90.2 | 93.3 | 92.1 | 90.0 | 89.7 | 87.7 | 86.5 | 87.1 | 87.2 |
| ResNet-50 | 98.6 | 98.2 | 99.4 | 98.3 | 98.7 | 97.6 | 98.5 | 97.4 | 92.3 | 92.2 | 92.6 | 91.1 | 90.9 | 89.9 | 90.7 | 89.1 |
| DenseNet-121 | 98.1 | 97.9 | 99.2 | 97.4 | 99.1 | 99.1 | 97.6 | 96.6 | 93.6 | 93.4 | 93.1 | 91.4 | 91.8 | 90.0 | 91.0 | 89.9 |
| MobileNetV3 | 98.7 | 97.2 | 98.4 | 97.6 | 97.9 | 96.3 | 97.8 | 97.4 | 93.0 | 92.1 | 93.4 | 91.8 | 91.6 | 89.2 | 90.1 | 90.7 |
| EfficientNet-B4 | 97.4 | 97.3 | 97.9 | 97.8 | 97.7 | 96.2 | 96.4 | 96.3 | 92.0 | 90.1 | 93.3 | 91.3 | 90.6 | 87.0 | 86.0 | 85.8 |
| RegNetY-800mf | 97.6 | 97.2 | 99.2 | 99.2 | 97.9 | 97.8 | 97.6 | 97.7 | 92.3 | 91.9 | 93.9 | 93.3 | 91.1 | 90.7 | 91.2 | 91.0 |
| ViT-B | 91.7 | 89.9 | 89.9 | 87.9 | 88.6 | 86.0 | 85.5 | 83.0 | 66.9 | 64.8 | 69.1 | 64.7 | 78.6 | 74.2 | 82.0 | 79.3 |
| Swin-T | 98.0 | 98.1 | 98.7 | 98.3 | 99.3 | 99.2 | 92.9 | 91.0 | 94.0 | 93.3 | 93.0 | 92.1 | 91.2 | 91.3 | 91.2 | 92.0 |
| WaveMix | 93.0 | 92.0 | 94.7 | 93.2 | 91.9 | 90.2 | 90.9 | 89.9 | 81.0 | 78.9 | 83.3 | 83.4 | 86.6 | 86.2 | 81.9 | 81.8 |
| ConvNeXt-Base | 98.9 | 98.7 | 99.2 | 99.1 | 97.9 | 97.0 | 97.6 | 97.4 | 93.9 | 93.3 | 93.6 | 92.2 | 90.0 | 90.1 | 91.7 | 90.0 |
| **WaveConvX** | **99.6** | **99.5** | **99.6** | **99.5** | **99.5** | **99.3** | **99.3** | **99.0** | **94.7** | **95.9** | **94.9** | **94.4** | **91.8** | **91.7** | **92.0** | **92.1** |

TABLE II: Generalization performance across datasets.

| Model | BreakHis 40× | | KBSMC | | LC25000 | |
|---|---|---|---|---|---|---|
| | Acc | F1 | Acc | F1 | Acc | F1 |
| VGG16 | 93.3 | 92.1 | 70.1 | 57.2 | 85.0 | 84.1 |
| ResNet-50 | 92.3 | 92.2 | 77.2 | 66.3 | 94.7 | 94.2 |
| DenseNet-121 | 93.6 | 93.4 | 80.0 | 67.2 | 98.7 | 98.6 |
| MobileNetV3 | 93.0 | 92.1 | 78.7 | 66.1 | 98.2 | 97.1 |
| EfficientNet-B4 | 92.0 | 90.1 | 80.1 | 67.1 | 96.5 | 96.6 |
| RegNetY-800mf | 92.4 | 91.9 | 81.4 | 72.2 | 94.7 | 94.2 |
| ViT-B | 66.9 | 64.8 | 70.9 | 59.9 | 95.0 | 95.0 |
| Swin-T | 94.0 | 93.3 | 80.2 | 69.7 | 85.9 | 84.4 |
| WaveMix | 81.0 | 78.9 | 58.8 | 46.4 | 92.3 | 92.1 |
| ConvNeXt-Base | 93.9 | 93.3 | 80.5 | 68.8 | 93.8 | 92.7 |
| **WaveConvX** | **94.7** | **95.9** | **83.3** | **74.4** | **99.3** | **99.2** |

with performance margins of 0.4%–2.0% over ConvNeXt-Base and ResNet-50. For the more challenging eight-class classification, WaveConvX maintains F1-scores of 91.7%–95.9%, surpassing Swin-T by 2.6% at 40× while exhibiting better stability across magnifications.

To assess generalization, we evaluated all models on KBSMC (gastric cancer) and LC25000 (lung/colon cancer) datasets (Table II). WaveConvX achieves superior performance on both: 83.3% accuracy / 74.4% F1 on KBSMC (exceeding RegNetY-800mf by 2.2% F1) and 99.3% accuracy / 99.2% F1 on LC25000 (outperforming DenseNet-121 by 0.6% F1).

### C. Ablation Studies

We quantified each architectural component's contribution through systematic ablation studies (Table III):

Results confirm that: (1) Two-level DWT (99.6%) outperforms single-level decomposition (97.5%), validating hierarchical frequency analysis for multi-scale feature capture; (2) Our specialized processing (APGConv+Attention) delivers substantial gains (99.6% vs. 94.7% with simple convolution);

TABLE III: Ablation Studies on BreakHis 40× binary classification.

| Factor | Setting | Accuracy (%) |
|---|---|---|
| Wavelet Family | sym4 / db2 / db4/ **Haar** | 96.2 / 97.8 / 98.4 / **99.6** |
| # Decomposition Levels | 1-level / **2-level** | 97.5 / **99.6** |
| Sub-Band Processing | Simple Conv / **APGConv+Attn** | 94.7 / **99.6** |
| Module Placement | After Stage 1 / 2 / **3** | 96.8 / 98.3 / **99.6** |
| Data Augmentation | Minimal / Moderate/ **Heavy** | 98.2 / 99.3 / **99.6** |

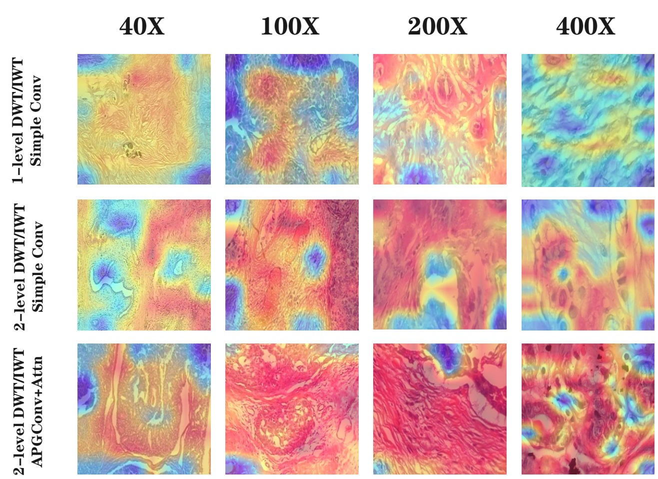


Fig. 4: Grad-CAM visualization of frequency-specific attention patterns.

(3) Optimal module placement occurs after Stage 3 (99.6%), balancing feature abstraction with frequency enhancement; (4) Haar wavelets (99.6%) outperform alternatives (96.2%-98.4%) due to computational efficiency; and (5) Robust data augmentation proves crucial for generalization (99.6% vs. 98.2%).

Figure 4 demonstrate that 2-level DWT/IWT combined with APGConv+Attn achieves superior localization accuracy across 40X–400X magnifications, with distinctively deeper and more concentrated attention maps compared to 1-level



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DWT/IWT and Simple Conv, validating their effectiveness.

## V. Discussion

### A. Computational Cost

Each DWT/IDWT is a lightweight $2\times2$ convolution or transpose convolution with fixed (or minimally learnable) weights. The sub-band processing adds modest overhead in Adaptive Power Gabor and attention blocks. Overall, training time increases by roughly 10–15% compared to a plain ConvNeXt, a reasonable cost for the accuracy gains.

### B. Limitations and Future Work

We focus on classification. Future work could extend wavelet-based enhancement to detection and segmentation tasks, such as localizing malignant regions. Additional medical modalities like CT or MRI may benefit from different wavelet families or deeper decomposition. Exploring synergy with full transformer blocks in wavelet space is also an exciting avenue.

## VI. Conclusion

In the paper, we propose a **WaveConvX** with improved multi-level wavelet enhanced ConvNeXt network for cancer pathological image classification. Experiments on BreakHis, KBSMC, and LC25000 show that our proposed WaveConvX significantly outperforms several compared mainstream baselines and SOTA methods. By introducing Adaptive Power Gabor Convolution (APGConv) alongside an attention mechanism, our model effectively preserves discriminative features in the frequency domain. In all, our work demonstrates the significant potential of wavelet-enhanced CNNs in achieving increasing performance for histopathological image analysis, which is beneficial for paving the way for more robust and clinically applicable diagnostic tools.

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## References

[1] J. Ferlay, M. Colombet, I. Soerjomataram, C. Mathers, D. M. Parkin, M. Piñeros, A. Znaor, and F. Bray, "Estimating the global cancer incidence and mortality in 2018: GLOBOCAN sources and methods," *Int J Cancer*, vol. 144, pp. 1941–1953, Apr. 2019. Epub 2018 Dec 6. PMID: 30350310.

[2] A. N. Giaquinto, H. Sung, K. D. Miller, J. L. Kramer, L. A. Newman, A. Minihan, A. Jemal, and R. L. Siegel, "Breast cancer statistics, 2022," *CA Cancer J Clin*, vol. 72, pp. 524–541, Nov. 2022. Epub 2022 Oct 3. PMID: 36190501.

[3] G. Litjens, T. Kooi, B. E. Bejnordi, A. A. A. Setio, F. Ciompi, M. Ghafoorian, J. A. W. M. van der Laak, B. van Ginneken, and C. I. Sánchez, "A survey on deep learning in medical image analysis," *Med Image Anal*, vol. 42, pp. 60–88, Dec. 2017. Epub 2017 Jul 26. PMID: 28778026.

[4] M. N. Gurcan, L. E. Boucheron, A. Can, A. Madabhushi, N. M. Rajpoot, and B. Yener, "Histopathological image analysis: A review," *IEEE Rev Biomed Eng*, vol. 2, pp. 147–171, 2009. Epub 2009 Oct 30. PMID: 20671804; PMCID: PMC2910932.

[5] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, p. 84–90, May 2017.

[6] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2015.

[7] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Nature*, vol. 521, pp. 436–444, 2015.

[8] R. Deng, C. Cui, L. W. Remedios, S. Bao, R. M. Womick, S. Chiron, J. Li, J. T. Roland, K. S. Lau, Q. Liu, K. T. Wilson, Y. Wang, L. A. Coburn, B. A. Landman, and Y. Huo, "Cross-scale multi-instance learning for pathological image diagnosis," *Medical Image Analysis*, vol. 94, p. 103124, 2024.

[9] I. Daubechies, *Ten Lectures on Wavelets*. Society for Industrial and Applied Mathematics, 1992.

[10] S. Mallat, "A theory for multiresolution signal decomposition: the wavelet representation," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 11, no. 7, pp. 674–693, 1989.

[11] *The Wavelet Transform in Signal and Image Processing*, pp. 159–220. London: Springer London, 2007.

[12] S. Li, Y. Liu, X. Sui, C. Chen, G. Tjio, D. S. W. Ting, and R. S. M. Goh, "Multi-instance multi-scale cnn for medical image classification," 2019.

[13] Y. Yan, R. Lu, J. Sun, J. Zhang, and Q. Zhang, "Breast cancer histopathology image classification using transformer with discrete wavelet transform," *Medical Engineering Physics*, vol. 138, p. 104317, 2025.

[14] Y. Qi, X. Liu, H. Li, M. Liu, and W. Li, "Discrete wavelet transform-based cnn for breast cancer classification from histopathology images," in *Machine Learning for Cyber Security* (Y. Xu, H. Yan, H. Teng, J. Cai, and J. Li, eds.), (Cham), pp. 326–340, Springer Nature Switzerland, 2023.

[15] M. Unser, "Texture classification and segmentation using wavelet frames," *IEEE Transactions on Image Processing*, vol. 4, no. 11, pp. 1549–1560, 1995.

[16] L. Li, J. Wu, and X. Jin, "Cnn denoising for medical image based on wavelet domain," in *2018 9th International Conference on Information Technology in Medicine and Education (ITME)*, pp. 105–109, 2018.

[17] C. Burrus, R. Gopinath, and H. Guo, "Introduction to wavelets and wavelet transform—a primer," *Recherche*, vol. 67, 01 1998.

[18] F. A. Spanhol, L. S. Oliveira, C. Petitjean, and L. Heutte, "A dataset for breast cancer histopathological image classification," *IEEE Transactions on Biomedical Engineering*, vol. 63, no. 7, pp. 1455–1462, 2016.

[19] F. A. Spanhol, L. S. Oliveira, P. R. Cavalin, C. Petitjean, and L. Heutte, "Deep features for breast cancer histopathological image classification," in *2017 IEEE International Conference on Systems, Man, and Cybernetics (SMC)*, pp. 1868–1873, 2017.

[20] G. Campanella, M. G. Hanna, L. Geneslaw, *et al.*, "Clinical-grade computational pathology using weakly supervised deep learning on whole slide images," *Nat Med*, vol. 25, pp. 1301–1309, 2019.

[21] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, J. Uszkoreit, and N. Houlsby, "An image is worth 16x16 words: Transformers for image recognition at scale," 2021.

[22] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin transformer: Hierarchical vision transformer using shifted windows," 2021.

[23] Z. Liu, H. Mao, C.-Y. Wu, C. Feichtenhofer, T. Darrell, and S. Xie, "A convnet for the 2020s," in *2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 11966–11976, 2022.

[24] P. Jeevan and A. Sethi, "Wavemix: Resource-efficient token mixing for images," 2022.

[25] Y. Jeong, J. Chu, J. Kang, S. Baek, J.-H. Lee, D.-S. Jung, W.-W. Kim, Y.-R. Kim, J. Kang, and I.-G. Do, "Application of transcriptome-based gene set featurization for machine learning model to predict the origin of metastatic cancer," *Current Issues in Molecular Biology*, vol. 46, no. 7, pp. 7291–7302, 2024.

[26] A. A. Borkowski, M. M. Bui, L. B. Thomas, C. P. Wilson, L. A. DeLand, and S. M. Mastorides, "Lung and colon cancer histopathological image dataset (lc25000)," *arXiv preprint arXiv:1912.12142*, 2019.



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